



TECHNICAL & COMMERCIAL PROPOSAL

Service Agreement for Comprehensive Chiller Maintenance Services

Public-Private Partnership (PPP) Schools Programme — 14 Schools

Proposal Reference	MES/QTN/2026/AMC-CHW-320
Date of Issue	07 September 2026
Revision	Rev. 00 — Original Submission
Client / Main Contractor	Elegancia MEP — a subsidiary of Estithmar Holding
For the Attention of	Mr. Suhaib Kunnikkal Muhammed, Mechanical Project Engineer
Copy to	Mr. Rajeev Xavichen
Client Enquiry	RFQ e-mail dated 05 September 2026, with attachment “Scope of Work – Chillers.docx” and Chiller Schedules dated 05 September 2026
Contract Period	01 December 2026 – 30 September 2028 (22 months)
Assets Covered	44 nos. Trane air-cooled screw chillers — 48,916 kW (13,908 TR) total installed capacity
Lump Sum Contract Value	QAR 1,500,000.00 (Qatari Riyals One Million Five Hundred Thousand Only)
Validity	60 calendar days from the date of issue



Date: 07 September 2026

Ref.: MES/QTN/2026/AMC-CHW-320

M/s Elegancia MEP

A subsidiary of Estithmar Holding

Kind Attention: Mr. Suhaib Kunnikkal Muhammed – Mechanical Project Engineer

Subject: Technical and Commercial Proposal – Service Agreement for Planned Preventive, Corrective and Reactive Maintenance of Chillers, PPP Schools Programme (14 Schools)

Dear Sir,

We thank you for your enquiry of 05 September 2026 and for the chiller schedules subsequently issued on the same date, which confirm that air handling units, fan coil units and chilled water pumps are outside the scope of this Agreement.

Metri Engineering Services W.L.L. is pleased to submit herewith our Technical and Commercial Proposal for the comprehensive maintenance of forty-four (44) Trane air-cooled screw chillers installed across the fourteen (14) PPP School projects, representing a total installed cooling capacity of 48,916 kW (13,908 TR) and a connected chiller electrical load of approximately 20,963 kW.

Our proposal has been prepared on a lump sum basis in accordance with your Scope of Work and covers all labour, supervision, transportation, tools, calibrated instruments, personal protective equipment, minor consumables, third-party oil and chilled water analyses, in-house vibration measurement and trending, out-of-hours working, and an unlimited 24/7 emergency call-out service with a guaranteed two-to-four hour site response.

We draw your attention in particular to Section 3 of this proposal, which sets out in explicit detail the boundary between minor consumables (fully included in our lump sum), major consumables and spare parts (excluded from supply, but with installation, commissioning and testing fully included). This distinction has been made deliberately transparent to avoid ambiguity during the contract term and to allow a like-for-like comparison of the offers you receive.

Section 5 contains the clarifications arising from our review of the Scope of Work. We ask that your attention be drawn in particular to the following four qualifications, which are integral to this offer:

- (i) All equipment under this Agreement is air-cooled with screw compressors. The Condenser Side and Control & Safeties PPM activities as written are drafted for water-cooled machines; our interpretation for air-cooled plant is set out at Clarification 5.2.
- (ii) Compressor oil replacement, oil filter elements and gasket kits are not included and carry no allowance. They will be quoted separately once required, strictly in accordance with the OEM recommendation for the machine concerned.
- (iii) Minor refrigerant top-up is included and capped at a maximum of one (1) cylinder per chiller for the whole Contract Period. Major top-ups and major leaks will be quoted separately.
- (iv) The chillers are to be handed over in good working condition before the commencement of the Agreement, following a joint condition assessment. Any repair or rectification arising from that assessment will be quoted separately.

We confirm that our commercial offer is submitted on the basis of quarterly payment in advance, as detailed in Section 4.4. In all other respects we accept your commercial terms.

We remain at your disposal for any technical or commercial clarification and look forward to the opportunity of serving Elegancia MEP on this landmark programme.

Yours faithfully,

Metri Engineering Services W.L.L.



1.1 Basis of this Proposal

This proposal has been prepared strictly on the basis of the following documents received from the Client:

- Request for Quotation e-mail from Elegancia MEP dated 05 September 2026, 12:11 hrs.
- Attachment: "Scope of Work – Chillers.docx".
- Chiller schedules issued by Elegancia MEP on 05 September 2026, 14:06 hrs, comprising four (4) Trane schedules covering the fourteen school projects.
- Client's written confirmation of 05 September 2026 that "AHU, FCU and pump are not included in this scope".

1.2 Asset Register – Equipment Covered

The Agreement covers the following chiller plant. All forty-four (44) units are Trane air-cooled packaged water chillers with screw compressors, operating on refrigerant R134a, selected at 13.4 °C entering / 6.7 °C leaving chilled water and 46 °C air-on condenser.

Important: All equipment under this Agreement is AIR-COOLED with screw compressors. There is no condenser water circuit, no cooling tower, no condenser water pump and no shell-and-tube condenser bundle on any unit. All condenser-side maintenance activities are therefore executed as air-cooled coil maintenance, as set out in Clarification 5.2, which forms an integral part of this offer.

School(s)	Units per School	Trane Model	Cooling Capacity (kW/unit)	Input Power (kW/unit)	CHW Flow (L/s per unit)	Configuration	Total Capacity per School (kW)
1, 2, 4, 5, 6, 7, 9, 10, 11, 12 (10 nos.)	3	RTAG375H_DS	1,100	473	39.10	2 Duty + 1 Standby	3,300
3 and 8 (2 nos.)	3	RTAG375H_DS	1,110	477	39.46	2 Duty + 1 Standby	3,330
13	4	RTAG410X	1,247	512	44.33	3 Duty + 1 Standby	4,988
14	4	RTAG375H	1,067	465.8	37.93	3 Duty + 1 Standby	4,268
TOTAL — 14 Schools	44 units	—	—	20,963 kW total	—	—	48,916 kW (13,908 TR)

1.3 Contract Period and Visit Structure

The Contract Period is 01 December 2026 to 30 September 2028 — twenty-two (22) calendar months — with eight (8) maintenance visits per school. This aligns with a quarterly maintenance cycle. In accordance with the Scope of Work, the eight visits per school are structured as follows:

Visit Type	No. per School	Scope Content	Frequency
Major Visit	2	Condenser Side PPM + Control & Safeties PPM + Lubrication System PPM + Motor & Starter PPM + full Compressor PPM, including annual oil analysis, chilled water sampling and analysis, and vibration measurement at all defined points	Once per contract year
Minor Visit	6	Compressor PPM — pressure and temperature verification, vibration check, control circuit testing, lubrication system check, operating log, condenser coil cleaning, deficiency reporting	Three times per contract year
Total per School	8	—	Quarterly cycle

Across the fourteen schools this equates to 112 scheduled school-visits and 352 individual chiller services over the Contract Period, in addition to the mobilisation, condition assessment, takeover attendance, training and unlimited emergency call-out obligations described below.



2. Technical Proposal

2.1 Maintenance Philosophy

The chiller plant serving these schools is new equipment being handed over from construction. The single greatest risk to plant life-cycle cost on a project of this nature is the first eighteen months of operation, during which installation-related defects, control mis-configuration, water-side contamination and commissioning-stage settings typically surface. Our approach is therefore built on three principles:

- Establish a defensible baseline. A full condition assessment and snagging report within two weeks of takeover fixes the condition of every machine at the point of transfer, protecting both parties and providing the datum against which all subsequent performance is judged.
- Maintain to OEM standard, not to minimum standard. All PPM activities follow Trane RTAG series service literature and recommended intervals, executed by technicians experienced on Trane screw chiller platforms and the Tracer™ controls family.
- Predict rather than react. Annual oil spectrographic analysis, chilled water chemical analysis and vibration trending of every compressor are used to identify bearing wear, moisture and acid ingress, and water-side fouling before they become failures. Trend data is reported quarterly with recommendations.

2.2 Major Visit — Detailed Scope (Twice per School over the Contract)

Each Major Visit is performed on every chiller at the school and comprises the following, in full compliance with the Client's Scope of Work:

2.2.1 Condenser Side PPM

- Full refrigerant leak survey of both circuits using calibrated electronic leak detection and, where required, UV tracer techniques; leak check results formally recorded and reported.
- Rectification of minor refrigerant leaks — flare and mechanical joints, valve packings, schrader cores, service port caps, minor braze repairs on accessible copper lines.
- Major leaks (microchannel/condenser coil body, evaporator shell, compressor casing, pressure vessel welds) inspected, quantified, photographed and reported for separate quotation.
- Chemical cleaning of the air-cooled condenser coil banks using low-pressure foaming coil cleaner and controlled rinse, followed by fin combing and restoration of fin geometry; verification of unobstructed air path and fan operation.
- Recording of chilled water entering and leaving temperatures and pressure drop across the cooler, direct comparison against the design values of 13.4 °C / 6.7 °C and the scheduled kPa limit, with deviation analysis.
- Chilled water sampling and full laboratory analysis (pH, conductivity, total dissolved solids, chlorides, sulphates, total hardness, total iron, total bacteria count), with written interpretation and corrective recommendations.
- Verification of refrigerant charge, subcooling and superheat; approach temperature calculation and fouling assessment.

2.2.2 Control and Safeties PPM

- Internal and external cleaning of the chiller control and starter panel; removal of dust and construction debris; verification of panel gasket and IP integrity.
- Thermographic and physical inspection of all wiring, terminations and busbar connections for tightness, overheating and discolouration; re-torquing of terminations to the manufacturer's specified values with a calibrated torque tool.
- Functional verification of all indicator lamps, alarm beacons, LED and LCD displays, and the Tracer™ operator interface.
- Testing of oil pressure safety devices, high and low pressure cut-outs, freeze protection, phase failure/reversal relay, flow proving interlock and all safety limits; recorded set-points versus OEM values.
- Verification of the chilled water pump starter auxiliary interlock signal at the chiller terminal interface (see clarification 5.2).

- Download, review and archiving of controller alarm history and fault logs; reset and reporting of latched alarms.

2.2.3 Lubrication System PPM

- Verification of oil heater operation, resistance and current draw during the off cycle, and confirmation of correct oil sump temperature.
- Oil level verification through the sight glass at stable operating conditions; minor oil top-up to correct the sight-glass level where required.
- Inspection and rectification of minor oil leaks at joints, sight glasses, filter housings and service connections.
- Oil sample collection under running conditions and full third-party laboratory analysis: viscosity, acid number, moisture content, wear-metal spectrography and particle count, with trend interpretation against the previous sample.
- Where the oil analysis indicates that an oil change is due, MES will report the finding with the laboratory evidence and submit a separate quotation for the compressor oil, oil filter and associated gasket kit strictly in accordance with the OEM (Trane) recommendation for the machine concerned. Compressor oil replacement, oil filter elements and gasket kits are NOT included in the lump sum and are not covered by any allowance — see Sections 3.2 and 3.3.

2.2.4 Motor and Starter PPM

- Inspection of all motor and starter wiring and terminations for tightness, overheating and discolouration.
- Inspection and rectification of contactor contacts for wear, pitting and welding; cleaning or replacement recommendation where beyond service limits.
- Inspection of all mechanical linkages for wear, security and correct clearances.
- Verification of the tightness of motor terminal connections to specified torque.
- Functional verification of all electrical interlocks and permissives.
- Measurement and recording of supply voltage on all three phases, confirmation within nominal $\pm 10\%$, together with phase current balance, and insulation resistance testing of the compressor motor windings.

2.3 Minor Visit — Detailed Scope (Six per School over the Contract)

Each Minor Visit is performed on every chiller at the school and comprises the Compressor PPM as specified:

- Measurement of suction pressure and temperature on both circuits and comparison against design; superheat calculation.
- Measurement of discharge pressure and temperature on both circuits and comparison against design; discharge superheat and compression ratio assessment.
- Vibration measurement of each compressor using MES's own calibrated vibration meter, at the six defined measurement points per compressor, with overall velocity recorded, trended and classified against ISO 10816 / ISO 20816 evaluation zones — refer to Section 2.3.1.
- General operational inspection and rectification of the unit, including fan motors, fan blades, guards, panels, isolators and mountings.
- Logging of all operating temperatures, pressures, voltages and amperages onto the standard MES service log sheet.
- Functional testing of the control circuit and staging/sequencing verification, including duty/standby rotation.
- Inspection and rectification of the lubrication system — oil level, oil pressure differential, heater function.
- Analysis of all recorded data against original design conditions, with deviation commentary.
- Review of operating procedures with the school operating personnel present on site.
- Chemical cleaning of the condenser coil banks.
- Issue of a written service report including the operation log and a clear statement of any uncorrected deficiencies detected.

2.3.1 Vibration Measurement — Instrumentation, Measurement Points and Acceptance Criteria

The chillers under this Agreement employ semi-hermetic screw compressors, in which the rotating elements and the drive motor are enclosed within a single sealed pressure housing and are not externally accessible. Vibration measurement on machines of this construction is therefore taken at the compressor casing and at the mounting, and the diagnostically significant parameter is the overall vibration velocity and its trend over time. MES carries out all vibration measurement in-house, using its own calibrated instruments, as set out below. This regime is included in the lump sum in full.

Instrumentation

- Handheld digital vibration meter with shear-type piezoelectric accelerometer and magnetic mounting base.
- Velocity measurement 0.1 – 199.9 mm/s RMS over the frequency band 10 Hz – 1 000 Hz, in accordance with the measurement band specified by ISO 10816 / ISO 20816 for machinery vibration evaluation.
- Acceleration measurement 0.1 – 199.9 m/s² peak, over 10 Hz – 1 kHz (low band) and 1 kHz – 15 kHz (high band), the latter providing a high-frequency condition indication.
- Displacement measurement 0.001 – 1.999 mm peak-to-peak.
- Measurement accuracy $\pm 5\% \pm 2$ digits.
- Each instrument is calibrated annually by an independent laboratory, with certificates traceable to national standards and available for the Client's inspection at any time. The cost of calibration is included in the lump sum.
- MES holds two (2) such instruments on this contract so that continuity of measurement is maintained while one instrument is away for calibration.

Measurement Points — Six (6) per Compressor

The following six measurement points are taken on every compressor, on every chiller, at every visit:

Point	Location	Direction
V1	Compressor casing — motor (suction) end	Horizontal
V2	Compressor casing — motor (suction) end	Vertical
V3	Compressor casing — rotor (discharge) end	Horizontal
V4	Compressor casing — rotor (discharge) end	Vertical
V5	Compressor casing — rotor (discharge) end	Axial
V6	Compressor mounting foot / base rail — isolator and structural looseness check	Vertical

Each measurement point is permanently marked on the machine during the initial condition assessment, so that every subsequent reading is taken at an identical location with the magnetic base. This is essential to the repeatability of the trend data. All readings are taken with the machine running at stable operating conditions.

Frequency and Recording

- All six points on every compressor are measured at every visit, Major and Minor — eight (8) sets of readings per compressor over the Contract Period.
- Condenser fan motors are checked by inspection at every visit, and vibration readings are taken on every fan motor at the annual Major Visit.
- Readings are recorded on the standard MES vibration log sheet, issued with the individual chiller service report, and entered into the vibration trend register maintained for each machine for the duration of the Contract.
- The trend register, with commentary on any change, is issued to the Client with the quarterly contract performance report.

2.4 Condition Assessment and Snagging Report

Within two (2) weeks of takeover of each school, MES will carry out a full condition assessment of every chiller and submit a formal snagging report. The assessment covers mechanical condition, refrigerant charge and leak status, electrical and control integrity, water-side condition, vibration and noise baseline, safety device verification, physical/structural condition of the unit and its mountings, and the completeness of OEM documentation, warranty certificates and as-built records. The report is issued in a defect register format with photographs, criticality ranking and recommended responsibility (construction contractor, OEM, or maintenance contractor).

2.5 Attendance at Testing, Commissioning and Takeover

MES will attend the T&C and takeover sessions conducted by the construction contractor at each of the fourteen schools, witnessing performance verification, safety device testing and control sequence demonstration, and will register any observations formally before acceptance. This attendance is included in the lump sum and is priced on the basis of attendance being called with not less than seven (7) days' notice.

2.6 Emergency and Reactive Maintenance

MES will operate a 24 hours / 7 days emergency response service for the full Contract Period with the following commitments:

- A single dedicated 24/7 hotline number and named escalation contacts (Contract Manager and Service Manager).
- Unlimited emergency call-outs at no additional charge, with attendance on site within two (2) to four (4) hours of the call being logged, day or night, including weekends and public holidays.
- Attendance, diagnosis, fault isolation, rectification where within the included scope, and restoration of duty/standby cooling capacity.
- A written incident report within twenty-four (24) hours of each attendance stating cause, action taken, parts required and preventive recommendation.
- Where the rectification requires spare parts or major consumables, these are quoted and supplied under Section 4.3; the attendance, diagnosis, labour, installation, commissioning and testing remain free of charge.
- All maintenance and rectification works are performed outside school operating hours where required, at no additional charge.

2.7 Operator Training

At each school MES will deliver a structured half-day familiarisation session for the site operations team covering daily operation and monitoring, interpretation of the Tracer™ interface and common alarms, first-line reactive actions, precautionary measures, safe isolation, log-sheet completion and escalation procedure. Training material and attendance records are provided; refresher sessions are delivered on request at no additional charge.

2.8 Resources and Organisation

Resource	Responsibility
Contract Manager	Overall contract governance, client interface, KPI reporting, escalation authority
Senior Service Supervisor (Chillers)	Technical leadership, major visit supervision, data trend analysis, root-cause investigation, OEM liaison
Chiller Service Teams (2 nos.)	Execution of major and minor PPM visits; each team comprising 1 senior technician + 1 technician + 1 helper
Emergency Response Team	24/7 rostered call-out cover, 2–4 hour response
QA/QC & Documentation Officer	Service report verification, analysis report compilation, asset register and planner maintenance
Vehicles & Equipment	Fully equipped mobile service vehicles with calibrated gauges, digital manifolds, electronic leak detectors, calibrated vibration meters (2 nos.), thermal imaging camera, insulation

Resource	Responsibility
	and loop testers, torque tools, coil cleaning equipment, recovery unit and vacuum pump

2.9 Deliverables and Reporting

Deliverable	Timing
Asset register and unique asset tagging of all 44 chillers	Within 4 weeks of award
Annual PPM planner per school, agreed with the Client	Within 4 weeks of award
Method statements, risk assessments and OEM-based PPM checklists	Prior to first visit
Condition assessment and snagging report per school	Within 2 weeks of takeover
Individual chiller service report and operating log per visit	Within 5 working days of each visit
Oil analysis, chilled water analysis and vibration trending reports	Within 10 working days of sampling
Emergency attendance / incident report	Within 24 hours of attendance
Quarterly contract performance report — PPM completion, KPI compliance, open defects, trend analysis, recommendations	Within 10 days of quarter end
Annual plant condition and energy performance review	Annually
Training records and attendance sheets	After each session

2.10 Health, Safety and Quality

- All works executed under approved method statements and job-specific risk assessments, with permit-to-work compliance to the Client's and the school's requirements.
- All technicians hold valid Qatari trade certification and safety induction; full PPE provided by MES at its own cost.
- Refrigerant handling in full compliance with Ministry of Environment and Qatar Civil Defence requirements; recovery, storage and documented reclamation — no venting to atmosphere under any circumstances.
- All measuring and test instruments calibrated annually with certificates traceable to national standards and available for Client inspection.
- Comprehensive third-party liability and workmen's compensation insurance maintained throughout the Contract Period.

3. Scope Boundary — Inclusions and Exclusions

This section is provided in deliberate detail. Its purpose is to remove all ambiguity as to what the lump sum price covers, so that no dispute arises during the Contract Period and so that the Client can compare tenders on a genuinely like-for-like basis.

3.1 Included in the Lump Sum Price

- All eight (8) scheduled maintenance visits per school — two Major Visits and six Minor Visits — executed on every chiller at the school as detailed in Sections 2.2 and 2.3.
- All direct and supervisory labour, including overtime, night working, weekend and public holiday working as required to work outside school hours.
- All supervision, engineering, contract management, planning and administration.
- All transportation, mobilisation and demobilisation to and from all fourteen sites.
- All hand tools, power tools, specialist tools, lifting tackle for routine maintenance, and all calibrated test and measuring instruments, including calibration costs.
- All personal protective equipment, safety equipment, barricading, signage and safety consumables.
- All minor consumables as itemised in Section 3.2 below.
- Chemical cleaning of air-cooled condenser coils at every visit, including cleaning chemicals and fin combing.
- Annual third-party laboratory oil analysis for every chiller — sampling, transport, laboratory fee and interpretation.
- Annual chilled water sampling and full laboratory chemical and bacteriological analysis per school — sampling, transport, laboratory fee and interpretation.
- Vibration measurement of every compressor at all six defined measurement points at every visit, using MES's own calibrated instruments — including the instruments, their annual independent calibration, data capture, trending and ISO zone classification.
- Full condition assessment and snagging report per school within two weeks of takeover.
- Attendance at all testing, commissioning and takeover sessions from the construction contractor.
- Operator familiarisation training for the site team at every school, including refresher sessions on request.
- Unlimited 24/7 emergency call-outs with 2–4 hour response — attendance, diagnosis, fault isolation and labour, at no additional charge, for the entire Contract Period.
- Labour, tools and consumables for the rectification of defects arising outside the DLP period, excluding only the supply of spare parts and major consumables.
- Installation, fitting, evacuation, charging, commissioning and testing of all spare parts and major consumables — irrespective of who supplies them, and at no additional labour charge.
- Rectification of minor refrigerant leaks and minor oil leaks, including minor R134a top-up capped at one (1) standard cylinder (13.6 kg nominal) per chiller over the entire Contract Period.
- All reporting, documentation, asset register, planners and quarterly performance reporting as listed in Section 2.9.
- All insurances, statutory compliance, employee visas, accommodation, transport and welfare costs.



3.2 Consumables and Parts — Definition Matrix

The Client's Scope of Work requires the Second Party to provide all consumables while excluding the supply of spare parts. Because "consumables" is capable of very wide interpretation on chiller plant of this size, MES defines the boundary explicitly as follows. This matrix forms part of our offer.

Category A — Minor Consumables: INCLUDED, unlimited, at MES cost

Item Group	Items Included
Coil and plant cleaning	Alkaline and acidic foaming coil cleaning chemicals, degreasers, neutralisers, rinse aids, cleaning cloths, rags, brushes, fin combs
Electrical	Contact cleaner spray, anti-oxidant/jointing compound, cable ties, insulation and self-amalgamating tape, heat-shrink sleeving, ferrules and small crimp lugs, cable glands, control fuses up to 10 A, indicator lamps, LED pilot lights, terminal markers
Mechanical	General purpose and bearing grease, penetrating oil, thread-lock compound, nuts, bolts, washers, screws, spring washers, panel gaskets and self-adhesive weather-strip, silicone and RTV sealant, PTFE tape
Refrigerant works	Oxygen-free nitrogen for pressure testing and purging, brazing rods, flux, leak detection soap solution, UV tracer dye, schrader cores and valve caps, flare gaskets and copper washers, vacuum pump oil
Analysis	Oil and water sample bottles, sample tubing, labels, sample transport and courier to laboratory
Refrigerant — minor top-up allowance	Minor R134a top-up following rectification of a minor leak, capped at a maximum of one (1) standard cylinder (13.6 kg nominal) per chiller over the entire Contract Period. Consumption beyond this cap is excluded and quoted separately.
Compressor oil — minor top-up only	Minor top-up of compressor oil solely to correct the sight-glass level during a PPM visit. NOTE: compressor oil replacement, oil filter elements and gasket kits are expressly EXCLUDED and carry no allowance — see Category B below.
Safety and admin	All PPE, gloves, goggles, respirators, harnesses, barricade tape, warning signage, log sheets, report printing and data media

Category B — Major Consumables: EXCLUDED from supply | Installation and testing INCLUDED

These items are NOT included in the lump sum and carry no allowance. They are supplied either by the Client, or by MES against a separate quotation submitted for the Client's written approval at the time the requirement arises, priced at net cost plus 15 % handling and strictly in accordance with the OEM (Trane) recommendation for the machine concerned. In every case MES carries out the fitting, evacuation, charging, commissioning and testing free of charge.

Item Group	Items Excluded from Supply
Compressor oil replacement	Trane-approved POE compressor oil for any oil change or partial/full charge replacement. Quoted separately once required, on the basis of the oil analysis result and the OEM recommendation. No allowance is included.
Oil filters and gasket kits	Compressor oil filter elements and cartridges, oil filter housing gaskets and complete OEM gasket / seal kits associated with any oil change or component opening. Quoted separately once required, as per OEM recommendation. No allowance is included.
Refrigerant — major top-up	R134a in any quantity exceeding the one (1) cylinder per chiller minor top-up cap, and all recharge following a major leak, major leak repair, component replacement or system evacuation. Quoted separately once required.
Filtration — refrigerant circuit	Suction and liquid line filter drier cores and shells, and control panel ventilation filters
Water side	Chilled water treatment chemicals, corrosion inhibitors, biocides, glycol, strainer baskets and screens, chemical dosing equipment
Consumable hardware	Fan motor bearings, vibration isolator pads and springs, flexible connections, sacrificial anodes, large-section cable and busbar links

Category C — Spare Parts: EXCLUDED from supply | Installation, commissioning and testing INCLUDED

Per Clause 10 of the Client's Scope of Work, the supply of spare parts is excluded while their installation and testing are included. MES will supply parts against a separate approved quotation at net cost plus 15 % handling, or install Client-free-issued parts at no charge. Indicative list, not exhaustive:

Item Group	Items Excluded from Supply
Rotating equipment	Screw compressors and compressor overhaul kits, compressor motors, oil pumps, oil separators, condenser fan motors, fan blades and fan assemblies
Heat exchangers	Microchannel or fin-and-tube condenser coil sections, evaporator shell-and-tube barrels, and all associated re-piping
Valves and refrigerant devices	Electronic and thermostatic expansion valves, solenoid valves, service and isolation valves, pressure relief valves, check valves, liquid line sight glasses, oil level regulators
Instrumentation	Pressure and temperature transducers, thermistors, sensors, flow switches, differential pressure switches, safety pressure switches, oil pressure switches
Electrical and control	Contactors, overload relays, circuit breakers, MCCBs, soft starters and variable frequency drives, power factor capacitors, phase monitors, transformers, control relays, current transformers
Controls and software	Tracer™ UC800 / AdaptiView™ controllers, display modules, expansion and I/O boards, communication modules, licensed OEM firmware and software
Heaters and ancillaries	Compressor oil heaters and thermostats, crankcase heaters, panel heaters and anti-condensation devices

3.3 Exclusions

The following are expressly excluded from this proposal and from the lump sum price:

- Air handling units, fan coil units, chilled water pumps, cooling towers, chilled water pipework, valves, strainers, expansion tanks, pressurisation units, thermal insulation, ductwork, VAV terminals and BMS — confirmed excluded by the Client on 05 September 2026.
- Supply of all spare parts and major consumables as defined in Sections 3.2 Category B and C.
- Compressor oil replacement, oil filter elements and gasket / seal kits. These are excluded in their entirety, carry no allowance, and shall be quoted separately once required and strictly as per the OEM (Trane) recommendation for the machine concerned.
- Refrigerant R134a in excess of the one (1) cylinder per chiller minor top-up cap, and all major refrigerant top-ups, major leak investigation and major leak repair. These shall be quoted separately once required.
- Any repair, rectification, replacement or making-good arising from the joint pre-contract condition assessment. All such items shall be quoted separately for the Client's decision — see Section 5.4.
- Chilled water system flushing, passivation, chemical treatment, dosing and make-up water.
- Electrical works upstream of the chiller local isolator, including incoming cabling, distribution boards, MCCs and their maintenance.
- Any works falling within the Defects Liability Period obligations of the installing contractor or the equipment manufacturer's warranty. MES will diagnose, report and coordinate such claims but will not bear the cost of their rectification.
- Trane OEM specialist attendance, proprietary service software, controller reprogramming, firmware upgrades and licence fees — offered as a provisional item under Section 4.3.
- Rectification of design deficiencies, installation defects, non-compliant workmanship or capacity shortfalls attributable to the original design or installation.
- Major refrigerant leak repair requiring coil, vessel or pressure-boundary weld repair; full system evacuation, dehydration and recharge.

- Compressor overhaul, motor rewinding, coil re-coring, and any workshop repair works.
- Scaffolding, cranes, forklifts, mobile lifting equipment, riggers and structural works required for the removal or replacement of major components.
- Provision of temporary or standby cooling, chiller rental and associated connections. Provision of lifting equipment.
- Repair of damage caused by misuse, operator error, unauthorised intervention, vandalism, theft, electrical supply irregularities, water quality outside acceptable limits, sandstorm or other force majeure event.
- Civil, builder's, painting and structural works; repair or replacement of plant room doors, louvres, drainage and access arrangements.
- Statutory approvals, permits, KAHRAMAA and Civil Defence liaison, and any associated fees.
- Electricity, water, lighting, site storage, site office and welfare facilities, which shall be provided free of charge by the Client at each school.
- Performance bond, advance payment guarantee and any bank charges, unless separately agreed and priced.
- All taxes and duties not applicable at the date of this offer, and any newly introduced levy including VAT.
- Any Crane or special lifting equipment. Any Civil or modification works.
- Any item, activity or service not expressly stated as included in Section 3.1.



4. Commercial Proposal

4.1 Schedule of Prices

All prices are in Qatari Riyals (QAR), firm and fixed for the full Contract Period, with no escalation.

Item	Description	Qty (Schools)
1.0	Mobilisation, asset registration and tagging, full condition assessment and snagging report per school (within 2 weeks of takeover)	14
2.0	Attendance at Testing & Commissioning and takeover sessions from the construction contractor	14
3.0	Operator familiarisation training for site teams, including material and records	14
4.1	Planned Preventive Maintenance — Schools 1, 2, 4, 5, 6, 7, 9, 10, 11 & 12 (3 nos. RTAG375H_DS, 1,100 kW each) — 8 visits per school	10
4.2	Planned Preventive Maintenance — Schools 3 & 8 (3 nos. RTAG375H_DS, 1,110 kW each) — 8 visits per school	2
4.3	Planned Preventive Maintenance — School 13 (4 nos. RTAG410X, 1,247 kW each) — 8 visits	1
4.4	Planned Preventive Maintenance — School 14 (4 nos. RTAG375H, 1,067 kW each) — 8 visits	1
5.0	24/7 emergency and reactive maintenance — unlimited call-outs, 2–4 hour response, out-of-hours working, for the full 22-month Contract Period	14
TOTAL LUMP SUM CONTRACT VALUE		

Total Contract Value: QAR 1,500,000.00 — Qatari Riyals One Million Five Hundred Thousand Only.

The consolidated award discount is offered strictly on the basis that all fourteen schools are awarded to MES under a single Service Agreement. Should the award be partial, the sub-total rates at Item 1.0 to 5.0 shall apply without discount.

4.2 Schedule of Rates — Provisional and Optional Items

The following are excluded from the lump sum and shall be executed only against the Client's written instruction, valued at the following rates. In all cases the associated labour, installation, commissioning and testing are already included in the lump sum and are not re-charged.

Item	Description	Rate (QAR)
P1	Supply of refrigerant R134a in excess of the one (1) cylinder per chiller minor top-up cap; major top-up following major leak, repair or evacuation	Quoted separately once required (indicative current market rate QAR 110.00 per kg)
P2	Compressor oil replacement — Trane-approved POE oil, any quantity	Quoted separately once required, as per OEM recommendation
P3	Compressor oil filter elements / cartridges and associated OEM gasket and seal kits	Quoted separately once required, as per OEM recommendation
P4	Complete oil change package per chiller — oil, oil filter and gasket kit. MES labour, evacuation, charging, commissioning and testing remain free of charge.	Materials quoted separately per P2 and P3; labour included in lump sum
P5	Refrigerant recovery, evacuation, dehydration and recharge following major repair — per circuit, excluding refrigerant	4,500.00 per circuit
P6	Supply of spare parts as defined in Section 3.2 Category C	Net cost + 15 % handling
P7	Trane OEM specialist attendance for controller reprogramming, firmware upgrade or proprietary software intervention (post-DLP)	At cost + 10 % — Provisional Sum QAR 60,000
P8	Major leak investigation and repair — condenser coil, evaporator barrel or pressure vessel	Quoted case by case
P9	Compressor overhaul, motor rewinding, coil re-coring and workshop repairs	Quoted case by case
P10	Chilled water chemical treatment, dosing and analysis programme (annual, per school)	Quoted on request
P11	Additional PPM visit beyond the eight contracted	Per Section 4.2 visit rates
P12	Scaffolding, crane and lifting equipment for major component replacement	At cost + 10 %

4.4 Payment Terms

Notwithstanding Clause (b) of the Client's Scope of Work, MES's offer is submitted on the following payment basis, which reflects the front-loaded resourcing, standby manning and consumable stock-holding required to sustain a 24/7 unlimited-response obligation across fourteen sites:

Instalment	Trigger	Amount (QAR)
Instalment 1	Upon award / signature of the Service Agreement, in advance of the first maintenance cycle	187,500
Instalments 2 to 8	In advance, at the commencement of each subsequent quarterly maintenance cycle	187,500 each
Total — 8 instalments		1,500,000

- Payment terms: quarterly in advance, eight (8) equal instalments of QAR 187,500 each, aligned to the eight contracted maintenance cycles.
- Each invoice shall be raised at the commencement of the relevant quarterly cycle and shall be payable within fifteen (15) days of submission.

- Maintenance reports for the preceding cycle shall be submitted with the following invoice, satisfying the Client's reporting requirement.
- Provisional and optional works under Section 4.3 shall be invoiced separately on completion, payable within thirty (30) days of an approved invoice.
- Late payment beyond the agreed terms entitles MES to suspend non-emergency services after fourteen (14) days' written notice; emergency response will nevertheless be maintained.
- All prices exclude any tax, VAT or levy that may be introduced during the Contract Period; any such charge shall be added to the invoice at the prevailing rate.

4.5 Commercial Conditions

- Currency: Qatari Riyals (QAR). Prices are firm and fixed for the full Contract Period with no escalation.
- Validity of offer: 60 calendar days from the date of issue.
- Award: this offer is priced for the award of all fourteen (14) schools under a single Service Agreement.
- Commencement: 01 December 2026, subject to award and to handover of the plant in a condition consistent with Section 5.4.
- Should any school not be handed over by 01 December 2026, the corresponding visits shall be re-programmed and the contract value adjusted pro rata at the visit rates of Section 4.2.
- Performance bond, if required, at 5 % of contract value — cost to be added to the contract price.
- This proposal supersedes all prior correspondence and is subject to the mutual execution of a formal Service Agreement incorporating the clarifications at Section 5.

5. Clarifications

Our review of the enquiry documents has identified the following points, which we ask the Client to confirm in writing. They do not affect our price but are material to the correct execution of the Agreement.

5.1 School Grouping Discrepancy

The RFQ e-mail groups the schools for contract duration purposes as: Typical Schools (1, 2, 6, 7, 9, 10, 11, 12); Schools (4, 5, 13, 14); School 03; and School 08. The chiller schedules, however, group them technically as: Schools 1, 2, 4, 5, 6, 7, 9, 10, 11, 12 (typical, 3 units each); Schools 3 and 8 (3 units each); School 13 (4 units); and School 14 (4 units). Schools 4 and 5 appear in different groups in the two documents. As all four groups carry an identical contract period and an identical eight visits, this has no commercial effect and we have priced strictly against the technical grouping in the chiller schedules. Client confirmation is requested.

5.2 Air-Cooled Screw Chillers — Interpretation of the Condenser Side and Control PPM

The Scope of Work for the Condenser Side PPM and the Control & Safeties PPM is drafted for water-cooled machines. Every chiller scheduled for this Agreement is an AIR-COOLED packaged unit with screw compressors and an integral air-cooled condenser. There is no condenser water circuit, cooling tower, condenser water pump or condenser tube bundle anywhere in the scope. We have therefore interpreted these clauses as follows, and have priced strictly on this basis:

- “Cleaning of condenser” is executed as chemical cleaning of the air-cooled condenser coil banks with fin combing — performed at every visit, major and minor.
- “Check and record water temperature in and out” is applied to the evaporator (chilled water) circuit, recorded against the design 13.4 °C / 6.7 °C and the scheduled cooler pressure drop.
- “Collect water samples and conduct water analysis” is applied to the chilled water circuit, one representative sample per school per year. No condenser water circuit exists.
- Condenser tube cleaning, eddy-current tube testing and condenser water treatment are not applicable and are excluded.
- “Test the operation of the chilled water pump starter auxiliary contacts” is executed as verification of the pump interlock and flow-proving signal at the chiller terminal interface only. The pumps and their starters are outside this scope as confirmed by the Client.

5.3 Defects Liability Period

The chillers are new equipment and will be within their Defects Liability Period and manufacturer’s warranty at the commencement of this Agreement. Clause 10 of the Scope of Work requires MES to bear labour, tools and consumables for defects arising outside the DLP. We request that the DLP expiry date for each school be advised in writing at takeover. During the DLP, warranty parts, warranty labour and OEM obligations remain with the installing contractor and with Trane; MES will diagnose, document and coordinate all warranty claims at no charge, but shall not bear their cost. Our maintenance activity will be executed so as not to prejudice the manufacturer’s warranty.

5.4 Condition at Handover — Condition Precedent

Clause 4 of the Scope of Work requires the system to be handed over fully operational and free of defective parts. The following is a condition precedent to the commencement of the Maintenance Services at each school and forms an integral part of this offer:

- The chillers shall be handed over to MES in good working order, fully operational, fully commissioned and free of defects, BEFORE the commencement of the Maintenance Agreement at that school.
- A full condition assessment shall be conducted JOINTLY by MES, the Client and the construction contractor prior to takeover, and the resulting report shall be signed by all parties as the agreed record of plant condition at the point of transfer.
- Any repair, rectification, replacement, re-commissioning or making-good identified during that joint assessment is EXCLUDED from the lump sum. MES shall submit a separate quotation for such works for the Client’s decision; they

may be executed by MES upon written instruction, or by the construction contractor, or by the OEM under warranty, at the Client's election.

- The Maintenance Services shall not commence at a given school, and no maintenance visit shall be counted against the eight contracted visits, until the plant at that school has been accepted in accordance with the above.
- MES accepts no liability whatsoever for pre-existing defects, installation deficiencies, incomplete or non-compliant commissioning, capacity shortfalls, or missing OEM documentation identified in the joint assessment and not rectified prior to takeover. Where the Client elects to proceed with items open, they shall be listed as signed carried-over exceptions to the Agreement and shall remain the responsibility of the construction contractor or the OEM.

5.5 Access and Client-Provided Facilities

- Free, safe and unimpeded access to all plant areas at all times, including outside school hours, with security passes and gate clearance arranged by the Client for MES personnel and vehicles.
- Safe access to roof-level or elevated plant areas, including permanent ladders, walkways, guardrails and lighting, provided by the Client.
- Electricity, water and drainage at each plant area free of charge for maintenance and coil cleaning purposes.
- Availability of chiller operating logs, running-hour data and BMS trend data on request.
- The full set of OEM operation and maintenance manuals, as-built drawings, commissioning records, warranty certificates and Trane parts lists to be handed over at takeover.
- Confirmation of the location of all fourteen schools. Our 2–4 hour emergency response is offered on the basis of all sites being within mainland Qatar and accessible by road.

5.6 Refrigerant and OEM Controls

- Trane Tracer™ controller reprogramming, firmware modification and proprietary software intervention require OEM-authorized access. MES will procure and coordinate this attendance under Provisional Item P7.
- All refrigerant recovered during maintenance is handled, stored and reclaimed in accordance with Ministry of Environment requirements, with documented traceability. Refrigerant recovery cylinders and reclamation records are provided by MES.
- Refrigerant top-up beyond the included allowance is subject to prior written approval and is charged under Item P1. MES will not repeatedly top up a machine with an unrepaired major leak, and will formally report such conditions for the Client's instruction.

5.7 General

- MES reserves the right to substitute the sequence of major and minor visits within the annual cycle to suit school occupancy and seasonal load, subject to the Client's agreement to the annual planner.
- The eight (8) visits per school over a 22-month period represent a quarterly cycle. Confirmation is requested that no additional visits are expected within the Contract Period.
- MES shall not be liable for consequential loss, loss of use, loss of profit or business interruption howsoever arising.

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Service Agreement for the Maintenance of Chillers at PPP 14 No's of Schools.

COMPLIANCE

SI No	Scope of work	Compliance	Remarks
1.	Planned Preventive, Corrective and Reactive Maintenance as per best practice and OEM recommendations on a Quarterly basis, including the Oil/Water/Vibration analysis once per year.	COMPLY	8 visits per school (2 major + 6 minor). Oil and chilled water analysis by third-party laboratory once per year. Vibration measured in-house at every visit.
2.	Full condition assessment and snagging report to be carried out and submitted within two weeks.	COMPLY	Submitted within 2 weeks of takeover of each school.
3.	Attending T&C and takeover sessions from the construction contractor.	COMPLY	Minimum 7 days written notice required.
4.	The System must be handed over fully operational & healthy without any defective parts unless otherwise mentioned by the second party once handed over the system.	COMPLY	Joint inspection by Elegancia, MES and the construction contractor before start of the contract. Any repair or rectification found will be quoted separately.
5.	Unlimited Free Emergency Calls 24x7, Response for Emergency within 2-4 hours	COMPLY	
6.	Work might be required to be performed out of the school's working hours.	COMPLY	
7.	Any required OEM support or reprogramming for the panel is covered by the Second Party (after the DLP).	COMPLY	Trane OEM attendance arranged and coordinated by MES. Covered by provisional sum QAR 60,000 at cost plus 10%.
8.	Any required consumables, tools, PPE, and instruments to perform the activities must be provided by the Second Party.	COMPLY	Minor consumables, tools, PPE and instruments all included. Compressor oil, oil filter, gasket kit and refrigerant above the included limit are excluded — refer proposal Sec. 3.2.

9.	Providing a basic training for the site team of the daily operation, reactive maintenance, and precautionary measures.	COMPLY	
10.	Labor, tools, and consumable costs for defects outside the DLP period shall be covered, except for the supply of spare parts; however, the installation and testing of spare parts are included.	COMPLY	Labour, tools and consumables included. Supply of spare parts excluded. Installation and testing of spare parts included free of charge.
	Condenser Side PPM: (Major Visit Per Year)		
1.	Check & rectify the unit thoroughly for minor refrigerant leaks.	COMPLY	Refrigerant top-up included up to one (1) cylinder per chiller for the whole contract period.
2.	Inspect for leaks and report major leak check results.	COMPLY	Inspection and reporting included. Repair of major leaks quoted separately.
3.	Repair minor leaks as required.	COMPLY	
4.	Check and record water temperature in & out and compare it with the design.	COMPLY	Chillers are air cooled. Applied to the chilled water (evaporator) side and compared against design 13.4 °C / 6.7 °C.
5.	Cleaning of condenser.	COMPLY	Air cooled condenser coils chemically cleaned at every visit.
6.	Collect Water Samples and conduct water Analysis.	COMPLY	Chilled water sample, one per school per year, tested by third-party laboratory. No condenser water circuit on air cooled chillers.
	Control and Safeties PPM: (Major Visit Per Year)		
1.	Inspect the control panel for cleanliness.	COMPLY	
2.	Inspect wiring and connections for tightness and signs of overheating and discoloration.	COMPLY	
3.	Verify the working condition of all indicator/alarm lights and LED/LCDs.	COMPLY	
4.	Test oil pressure safety device (as required).	COMPLY	

5.	Test the operation of the chilled water pump starter auxiliary contacts, if applicable	COMPLY	Interlock signal checked at the chiller panel only. Pumps are not in our scope, as confirmed by Elegancia on 05 Sep 2026.
	Lubrication System PPM: (Major Visit Per Year)		
1.	Verify the operation of the oil heaters.	COMPLY	
2.	Check the oil level and top-up oil if required.	COMPLY	Compressor oil supply & filters is excluded
3.	Check and rectify minor oil leaks.	COMPLY	
4.	Collect oil samples and conduct oil analysis.	COMPLY	One sample per refrigerant circuit per year, tested by third-party laboratory.
5.	Replace oil, and oil filter if required.	COMPLY	Labour included free of charge. Oil, oil filter and gasket kit are excluded and will be quoted separately as per OEM recommendation.
	Motor and Starter PPM: (Major Visit Per Year)		
	Inspect wiring and connections for tightness and signs of overheating and discoloration.	COMPLY	
1.	Check & rectify the condition of the contacts for wear and pitting.	COMPLY	Replacement contactors are spare parts — supply excluded, installation included.
2.	Check & rectify all mechanical linkages for wear, security, and clearances.	COMPLY	
3.	Verify the tightness of the motor terminal connections.	COMPLY	
4.	Verify the operation of the electrical interlocks.	COMPLY	
5.	Measure voltage and record. Voltage should be nominal voltage $\pm 10\%$.	COMPLY	We measure, record and report. Incoming supply voltage comes from the KAHRAMAA network and is not in our control.
	Motor and Starter PPM: (Major Visit Per Year)		This section is repeated in the form. Understood as one requirement, carried out once per major visit.
	Inspect wiring and connections for tightness and signs of overheating and discoloration.	COMPLY	As above.

1.	Check & rectify the condition of the contacts for wear and pitting.	COMPLY	As above.
2.	Check & rectify all mechanical linkages for wear, security, and clearances.	COMPLY	As above.
3.	Verify the tightness of the motor terminal connections.	COMPLY	As above.
4.	Verify the operation of the electrical interlocks.	COMPLY	As above.
5.	Measure voltage and record. Voltage should be nominal voltage $\pm 10\%$.	COMPLY	As above.
	Compressor PPM: (Minor Visit 3 Times Per Year)		
1.	Check suction pressure and temperature and compare it with the design.	COMPLY	
2.	Check discharge pressure and temperature and compare it with the design.	COMPLY	
3.	Conduct vibration analysis for compressors.	COMPLY	Measured in-house using a calibrated vibration meter at 6 points per compressor, at every visit. Readings compared against ISO 10816 zones and trended. Refer proposal Sec. 2.3.1.
4.	Check & rectify the general operation of the unit.	COMPLY	Labour included. Supply of spare parts excluded.
5.	Log the operating temperatures, pressures, voltages, and amperages.	COMPLY	
6.	Check the operation of the control circuit.	COMPLY	
7.	Check & rectify the operation of the lubrication system.	COMPLY	
8.	Analyze the recorded data and compare the data with the original design conditions.	COMPLY	
9.	Review operating procedures with operating personnel.	COMPLY	

10.	Provide a written report of completed work, and operation log and indicate any uncorrected deficiencies detected.	COMPLY	Report issued within 5 working days of each visit.
11.	Cleaning of condenser.	COMPLY	Air cooled condenser coils cleaned at every visit.
	Contract Value and Payment Terms		
	The total Contract Value shall be QAR (Qatari Riyals), representing a lump-sum amount inclusive of all labor, tools, consumables, PPE, and other incidental costs.	COMPLY	QAR 1,500,000 lump sum, firm and fixed for the full contract period. Exclusions as per proposal Sec. 3.3.
	Payment Terms:		
	Invoices to be submitted with maintenance reports.	COMPLY	
	Payment within 30 days from receipt of an approved invoice.	NOT COMPLY	Payment quarterly in advance — 8 equal instalments of QAR 187,500 each, payable within 15 days of invoice.

	Contract Duration-Schools	Contract Period	Compliance
	Schools (Nos. 1, 2, 6, 7, 9, 10, 11, and 12) 8 Schools- with total number of visits for each school: 8	1 st December 2026 to 30 th September 2028 (2 Years)	COMPLY — note: 01 Dec 2026 to 30 Sep 2028 is 22 months, not 2 years. Priced for 8 visits per school.
	Schools (Nos. 4, 5, 13, and 14)- 4 Schools- with total number of visits for each school: 8	1 st December 2026 to 30 th September 2028 (2 years)	COMPLY
	School 03- with total number of visits: 8	1 st December 2026 to 30 th September 2028 (2 years)	COMPLY
	School 08- with total number of visits: 8	1 st December 2026 to 30 th September 2028 (2 years)	COMPLY

Sign & Stamp

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